

lead to overestimated stellar masses and underestimated values for X_c/X_{ini} . On the other hand, Avdeeva et al. (2024) show that the effective temperatures from GSP-Phot and the GALAH survey are globally in good agreement (limited to effective temperatures below 8000 K). Asteroseismic effective temperatures will be inferred from follow-up studies of this sample and may shed light on the systematics.

The distributions of stellar masses of the sample of g-mode pulsators are in general agreement with a Salpeter IMF, except that we find an excess of stars within the mass range of the γ Dor pulsators. This suggests that flux blocking at the bottom of the thin convective envelope is a highly effective excitation mechanism for g-mode oscillations (Guzik et al. 2000; Dupret et al. 2005). On the other hand, g-mode excitation by the κ -mechanism seems to occur effectively in stars covering masses from about $2 M_\odot$ to about $9 M_\odot$, but our sample is not representative for the stars with a mass above $4 M_\odot$. As long as we do not have the full lists of detected identified oscillation modes per star, we cannot make more detailed inferences on the effectivity of the excitation mechanisms in terms of particular modes. Such inferences will be the subject of further studies.

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Appendix A: Results for solar metallicity grid

In this appendix, we show the derived mass-luminosity relation, mass-radius relation, and X_c/X_{ini} -core-mass relation, when we assume a solar metallicity of $Z = 0.014$.

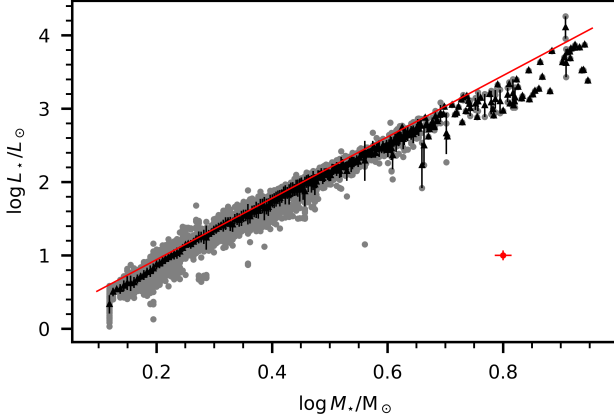


Fig. A.1: Same as Fig. 6, but for $Z = 0.014$.

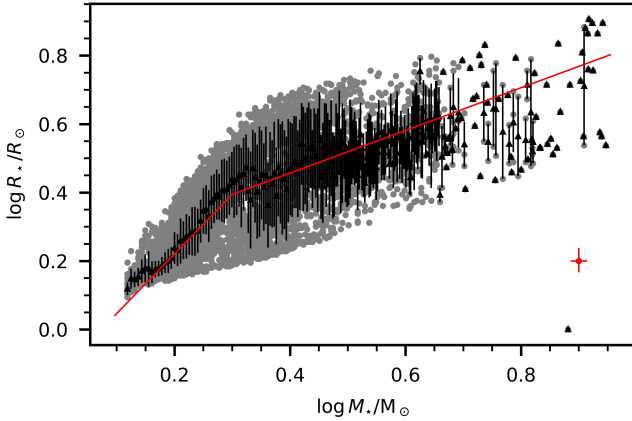


Fig. A.2: Same as Fig. 8, but for $Z = 0.014$.

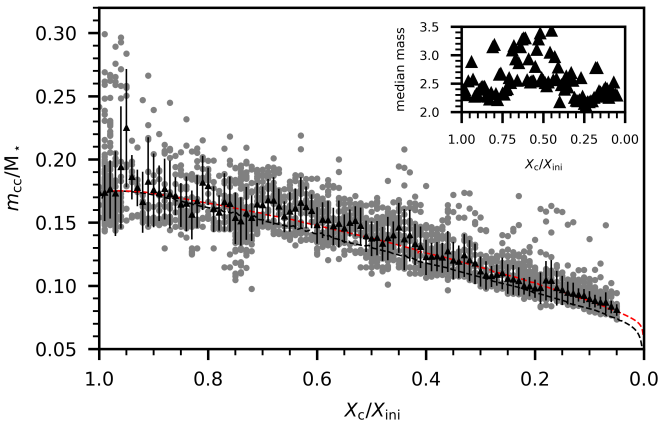


Fig. A.3: Same as Fig. 9, but for $Z = 0.014$.

Appendix B: Accuracy of the methodology

In this appendix, we discuss the accuracy of the methodology presented in this work. As a first test, we computed additional stellar evolution models that have not been used to train the CNF. Figure B.1 shows the points sampled from the CNF compared to the actual models. The errors on the interpolated models are small enough that we would have to at least increase the number of masses by a factor five and similar for values of f_{CBM} (see left panel of Fig. B.1 where the distance between two evolution models of different masses in grid are shown). Therefore, the relatively small error introduced by using a CNF is worth it.

As a second test, we sampled points on the MESA models and tried to recover the input parameters from the effective temperature and luminosity with the methodology described in Section 3. We assigned an uncertainty of 8 per cent on $\log L_*$ and 0.3 per cent on $\log T_{\text{eff}}$, representative of the typical uncertainties of the *Gaia* sample. Figure B.2 shows the difference in the recovered parameters compared to the input value. Assuming these uncertainties, we recover in ~ 75 per cent of the cases the actual convective core mass within the 68%-confidence interval, while for the other parameters in more than 90 per cent of the cases. In general, we do not observe any systematic offset in the discrepancies between the recovered values and actual values. Therefore, since we use a large sample of stars, these errors are mostly averaged out.

For the third test, we quantified the impact of assuming an incorrect metallicity in the modelling. We did this test for the most extreme case, that is, we modelled points from the $Z = 0.0045$ grid with the CNF that is trained on the $Z = 0.014$ grid. We note that we expect the majority of stars in the sample to have a metallicity between these two values (de Laverny et al. 2024). Figure B.3 shows the recovered stellar parameters. The same uncertainties as previously mentioned are assumed for the luminosity and the effective temperature. The recovered mass is on average about 15 per cent larger when an incorrect metallicity is assumed. For the fractional core masses, we find an average offset of roughly 0.02, as assuming a higher metallicity than in reality leads to finding younger stars. In general, the recovered radius is still correct, with some models for which the radius is overestimated.

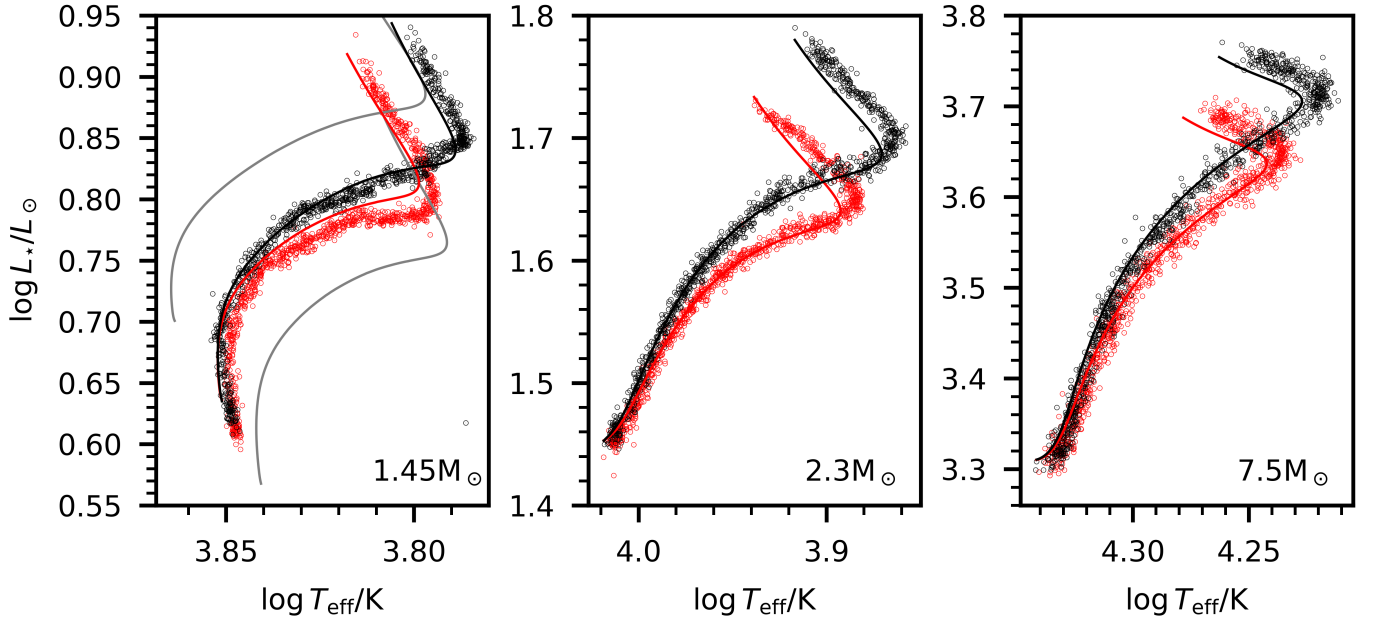


Fig. B.1: HRD showing stellar evolution tracks unseen by the CNF, and the sampling of these tracks by the CNF. The red colour indicates $f_{\text{CBM}} = 0.01$, the black colour indicates $f_{\text{CBM}} = 0.02$. In the left panel, tracks for 1.4 and 1.5 $M_{\odot}$ ($f_{\text{CBM}} = 0.015$, $\omega_0 = 0.05$) are shown in grey to illustrate the distance between two evolution tracks in the grid and the interpolation error of the CNF.

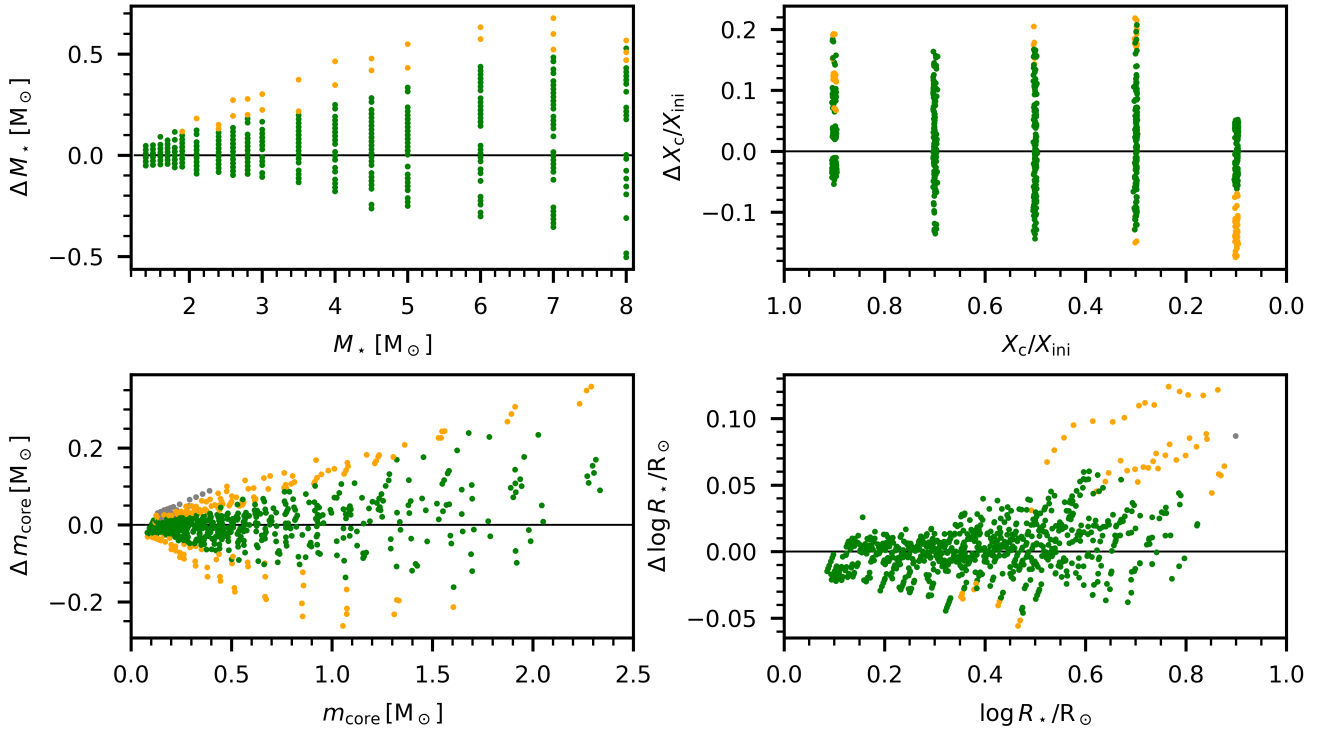


Fig. B.2: Discrepancy between the actual value and the recovered parameter for a set of validation data points ($\Delta x = x_{\text{actual}} - x_{\text{recovered}}$). The green colour indicates values that are recovered within the 68%-confidence interval, the orange colour within twice this interval.